Pre-analysis Plan for the #manydaughters Project

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In this Pre-analysis Plan (PAP), we briefly describe our strategy to test four hypotheses as stated by the principal investigators of the #manydaughters project (Albrecht et al., 2025) using SOEP data (Goebel et al., 2019).

All four hypotheses laid out by the PIs (Albrecht et al., 2025) are interested in the causal effect of having daughters versus not having daughters on different outcomes. We denote this treatment variable *D*. Unconditionally, with pregnancies as unit of analysis, we assume the assignment of sex at conception to be as good as random (Wang et al., 2025). Because certain parents might have preferences for the sex of a child or sex composition of their children, it is possible that parents self-select into having more children—thereby increasing the likelihood of belonging to the treatment group—conditional on the sex of their first child or sex composition of their children already born (Green et al., 2023; Oswald & Powdthavee, 2010; Rosenzweig & Wolpin, 2000). If and when parents who self-select into the treatment group differ on the respective outcome variable compared to parents that do not self-select in this way, the causal effect of interest would be biased. Because we are not willing to rule this pattern out by assumption, we are controlling for the number of children in all four models. To further minimize bias from selecting into parenthood—that is, having children at all—we restrict all models to parents. Respondents without children are, thus, excluded from the analysis.

To explain variation in the outcome variables—thus, increasing the precision of the estimated effect of *D*—we control for a small number of additional covariates, namely respondent's age (*bioage*), education in years (*pgbilzeit*), respondent's self-declared sex or gender (*sex*), and—when estimating pooled cross-section regression models with data from more than one point in time—for the year of data collection (*syear*).

H1

> Having daughters leads to stronger support for gender equality (as measured by attitudes).
>
> That is, denoting *X* as a measure of support for gender equality,
>
> (*X* for respondents with daughters) > (*X* for respondents without daughters)

Research has shown that men with daughters have more progressive gender role attitudes (Shafer & Malhotra, 2011) but there is considerable variation on how the daughter effect has been operationalized. While some argue that the gender of the first child is important (Lee & Conley, 2016; Shafer & Malhotra, 2011), others assert that the daughter effect occurs when individuals have at least one female offspring (Oswald & Powdthavee, 2010). Support for gender equality was measured by the mean value from a set of items relating to gender role attitudes which were only available in the 2012 wave of the SOEP (variables *plh0301, plh0302_v1, plh0307*, and reverse-coded versions of *plh0303, plh0308_v1*, and *plh0309*). Given that reversed causality should not be an issue with our

treatment variable and based on the assumption that, conditional on the number of children, children's sex composition, *D*, is as good as random, we identify the causal effect of *D* without obvious bias. We estimate the following OLS regression:

$$X_i = \beta_0 + \beta_1 D_i + \beta_2 F_i + \boldsymbol{\gamma C_i} + \epsilon_i,$$

where X_i is the value for individual i on our measure of support for gender equality, β_0 is an intercept, β_1 is the causal effect of D on X, D_i is the sex composition of the respondent's children with $D = 1$ if the respondent has a daughter and $D = 0$ otherwise, F_i is a dummy variable that equals 1 for self-reported women and 0 for self-reported men, $\boldsymbol{C_i}$ is a vector of controls, namely number of respondent's children, respondent's age, and education in years, and ϵ_i is a heteroskedasticity-robust error term. Hypothesis H1 states that $\beta_1 > 0$.

H2

Having daughters leads to more left-leaning political preferences.

That is, denoting *Y* as a measure of left-leaning political preferences,

(*Y* for respondents with daughters) > (*Y* for respondents without daughters).

Left-leaning political preferences could, in principle, be operationalized in numerous ways, including, e.g., partisanship, voting, and political beliefs. The prevailing view in political science and political psychology is that political orientations have at least a social and an economic dimension (Feldman & Johnston, 2014). Nevertheless, we operationalize (left-leaning) political preferences by taking respondents' self-placement on a simple left-right scale (variable *plh0004*, the scale runs from left, 0, to right, 10) (Kroh, 2007). First, this indicator is still one of the most widely used measures of political attitudes. Second, the SOEP hardly contains any longer item batteries for political issue positions that would allow for a more nuanced measurement. Because the left-right scale was fielded only in 2005, 2009, 2014, and 2019, we restrict our analysis to these years and estimate a pooled cross section via OLS of the following form:

$$Y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 F_{it} + \boldsymbol{\gamma C_{it}} + \boldsymbol{\delta T_{it}} + \epsilon_{it},$$

where Y_{it} is the value for individual i in survey year t on the left-right scale, β_0 is an intercept, β_1 is the causal effect of D on Y, D_{it} is the sex composition of the respondent's children with $D = 1$ if the respondent has a daughter and $D = 0$ otherwise, F_{it} is a dummy variable that equals 1 for self-reported women and 0 for self-reported men, $\boldsymbol{C_{it}}$ is a vector of controls, namely number of respondent's children, respondent's age, and education in years, $\boldsymbol{\delta T_{it}}$ is a vector of survey year fixed effects, and ϵ_{it} is a cluster-robust error term that accounts for repeated measures within individuals. Hypothesis H2 states that $\beta_1 < 0$.

H3

Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

That is, denoting *Z* as a measure of gender equality in the household,

(*Z* for respondents with daughters) > (*Z* for respondents without daughters)

Due to time constraints and to meet the submission deadline for this PAP, we keep this analysis very simple and approximate gender equality in the household in the following way: We operationalize gender equality in the household by constructing a variable that contains the mean number of hours spent on housework (*pli0043_h*) and childcare (*pli0044_h*) on a working day. We restrict the analysis to fathers. Conditional on the number of children, we interpret an increase in mean number of hours spent on housework and childcare by fathers as evidence for more gender equality in the household. To test H3, we estimate a pooled cross section via OLS of the following form:

$$Z_{it} = \beta_0 + \beta_1 D_{it} + \boldsymbol{\gamma C_{it}} + \boldsymbol{\delta T_{it}} + \epsilon_{it},$$

where Z_{it} is the value for individual i in survey year t on our measure of involvement in care work, β_0 is an intercept, β_1 is the causal effect of D on Z, D_{it} is the sex composition of the respondent's children with $D = 1$ if the respondent has a daughter and $D = 0$ otherwise, $\boldsymbol{C_{it}}$ is a vector of controls, namely number of respondent's children, respondent's age, and education in years, $\boldsymbol{\delta T_{it}}$ is a vector of survey year fixed effects (for years 2001 – 2020, reference category is 2000), and ϵ_{it} is a cluster-robust error term that accounts for repeated measures within individuals. Hypothesis H3 states that $\beta_1 > 0$.

H4

> Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes)
>
> That is, denoting X as a measure of support for gender equality,
>
> (X for male respondents with daughters − X for male respondents without daughters) > (X for female respondents with daughters − X for female respondents without daughters)

The hypothesis implies that the effect of having a daughter on the support for gender equality is stronger among fathers compared to mothers (Borrell-Porta et al., 2019). Thus, we operationalized this hypothesis by adding an interaction effect of having at least one daughter with respondent's gender to the model testing H1. All measures and the sample of analysis remain the same as described above for the model testing H1. To test H4 we estimate the following OLS regression:

$$X_i = \beta_0 + \beta_1 D_i + \beta_2 F_i + \beta_3 D_i F_i + \boldsymbol{\gamma C_i} + \epsilon_i$$

Where—in contrast to the model for H1—β_1 now captures the causal effect of D on X for men, while the causal effect of D on X for women is given by $\beta_1 + \beta_3$. Hypothesis H4 states that $\beta_3 < 0$.

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